Collecting Like Terms and Multiplying Terms

Cell Tuition 1-1 PRIVATE TUTORS

Name _____ Date _____ No calculator Version 1

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KEY WORDS like terms *(tūnga ōrite)* · term *(tūnga)* · coefficient *(tau whakarea)* · simplify *(whakamāmā)*

A Like terms with one letter ●○○

Remember: Like terms have exactly the same letters. Add or subtract the numbers in front: $3x + 5x = 8x$.

1. Simplify.

a) $3x + 5x$ = _____ **b)** $7a - 2a$ = _____ **c)** $4y + y$ = _____ **d)** $9b - 8b$ = _____

e) $2m + 3m + 4m$ = _____ **f)** $10p - 4p$ = _____ **g)** $6x - 6x$ = _____ **h)** $5n - 2n + 4n$ = _____

i) $k + k$ = _____ **j)** $8c - 3c - 2c$ = _____ **k)** $3d - 7d$ = _____ **l)** $-2x + 9x$ = _____

B Two kinds of term ●●○

Watch out: Only like terms can be collected. $3x + 2y$ cannot be made any simpler, and $4x + 5$ is not $9x$.

2. Simplify.

a) $3x + 2y + 4x$ = _____ **b)** $5a + 3 + 2a$ = _____ **c)** $6m + 4n - 2m$ = _____

d) $7p + 5 - 3p + 2$ = _____ **e)** $4x + 3y + 2x + 5y$ = _____ **f)** $8a + 6b - 3a - 2b$ = _____

g) $2c + 9 + 5c - 4$ = _____ **h)** $10x + 7y - 10x$ = _____ **i)** $5k + 1 + k + 6$ = _____

j) $9m - 2n - 4m + 5n$ = _____ **k)** $3a + 4b - 3b$ = _____ **l)** $6x - 8 - 2x + 3$ = _____

C Terms with powers and products ●●●

Remember: x^2 and x are not like terms. ab and ba are like terms, because $a \times b$ is the same as $b \times a$.

3. Simplify.

a) $3x^2 + 2x + 4x^2$ **b)** $2ab + 5ab$ **c)** $4a + 3ab - a$

d) $5y^2 - 2y^2 + 3y$ **e)** $6xy + 2x - 4xy$ **f)** $7n^2 + 3 - 5n^2 + 1$

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D Multiplying terms ●●○

> **Remember:** Multiply the numbers, then the letters: $3a \times 2b = 6ab$. A letter times itself gives a power: $a \times a = a^2$.

4. Simplify.

a) $3 \times 2a$ = _____ **b)** $4x \times 5$ = _____ **c)** $a \times b$ = _____ **d)** $3a \times 2b$ = _____

e) $x \times x$ = _____ **f)** $5m \times 4n$ = _____ **g)** $2y \times 3y$ = _____ **h)** $6p \times q$ = _____

i) $-2 \times 4x$ = _____ **j)** $3a \times (-5b)$ = _____ **k)** $2 \times 3a \times 4b$ = _____ **l)** $7x \times 2x$ = _____

E Dividing terms ●●●

> **Remember:** Divide the numbers, and cancel any letter that is on both the top and the bottom: $\frac{12a}{4} = 3a$ and $\frac{6ab}{2a} = 3b$.

5. Simplify.

a) $\frac{12a}{4}$ **b)** $\frac{10x}{5}$ **c)** $\frac{8y}{y}$ **d)** $\frac{6ab}{2a}$

e) $\frac{15m}{3m}$ **f)** $\frac{20xy}{4y}$ **g)** $\frac{9a^2}{3a}$ **h)** $\frac{18pq}{6pq}$

F Problems ●●●

6. Write an expression for each perimeter, then simplify it.

a) A triangle with sides $3x$, $4x$ and $5x$

b) A rectangle with length $5a$ and width $2a$

c) A rectangle with length $5x + 2$ and width $2x$

★ Extension

7. A rectangle is $3x$ long and $4x$ wide. Write a simplified expression for its area.

8. Simplify $5x + 3y - 2x + 4y - x - 6y$.

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Collecting Like Terms and Multiplying Terms

Cell Tuition 1-1 PRIVATE TUTORS

Name _____ Date _____ No calculator Version 2

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KEY WORDS like terms *(tūnga ōrite)* · term *(tūnga)* · coefficient *(tau whakarea)* · simplify *(whakamāmā)*

A Like terms with one letter ●○○

Remember: Like terms have exactly the same letters. Add or subtract the numbers in front: $3x + 5x = 8x$.

1. Simplify.

a) $4x + 3x$ = _____ **b)** $9a - 5a$ = _____ **c)** $6y + y$ = _____ **d)** $7b - 6b$ = _____

e) $3m + 2m + 5m$ = _____ **f)** $12p - 7p$ = _____ **g)** $8x - 8x$ = _____ **h)** $6n - 3n + 2n$ = _____

i) $h + h$ = _____ **j)** $9c - 4c - 3c$ = _____ **k)** $2d - 6d$ = _____ **l)** $-3x + 8x$ = _____

B Two kinds of term ●●○

Watch out: Only like terms can be collected. $3x + 2y$ cannot be made any simpler, and $4x + 5$ is not $9x$.

2. Simplify.

a) $2x + 3y + 5x$ = _____ **b)** $4a + 6 + 3a$ = _____ **c)** $7m + 2n - 3m$ = _____

d) $8p + 4 - 2p + 3$ = _____ **e)** $3x + 4y + 5x + 2y$ = _____ **f)** $9a + 5b - 4a - 3b$ = _____

g) $3c + 8 + 4c - 5$ = _____ **h)** $12x + 5y - 12x$ = _____ **i)** $4k + 1 + k + 7$ = _____

j) $8m - 3n - 5m + 6n$ = _____ **k)** $2a + 5b - 4b$ = _____ **l)** $7x - 9 - 3x + 4$ = _____

C Terms with powers and products ●●●

Remember: x^2 and x are not like terms. ab and ba are like terms, because $a \times b$ is the same as $b \times a$.

3. Simplify.

a) $2x^2 + 5x + 3x^2$ **b)** $4ab + 3ab$ **c)** $5a + 2ab - 2a$

d) $6y^2 - 4y^2 + 2y$ **e)** $7xy + 3x - 5xy$ **f)** $8n^2 + 2 - 6n^2 + 4$

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D Multiplying terms ●●○

> **Remember:** Multiply the numbers, then the letters: $3a \times 2b = 6ab$. A letter times itself gives a power: $a \times a = a^2$.

4. Simplify.

a) $4 \times 3a$ = _____ **b)** $2x \times 6$ = _____ **c)** $x \times y$ = _____ **d)** $4a \times 2b$ = _____

e) $y \times y$ = _____ **f)** $3m \times 6n$ = _____ **g)** $4y \times 2y$ = _____ **h)** $5p \times q$ = _____

i) $-3 \times 5x$ = _____ **j)** $2a \times (-7b)$ = _____ **k)** $3 \times 2a \times 5b$ = _____ **l)** $6x \times 3x$ = _____

E Dividing terms ●●●

> **Remember:** Divide the numbers, and cancel any letter that is on both the top and the bottom: $\frac{12a}{4} = 3a$ and $\frac{6ab}{2a} = 3b$.

5. Simplify.

a) $\frac{15a}{3}$ **b)** $\frac{12x}{6}$ **c)** $\frac{7y}{y}$ **d)** $\frac{8ab}{2a}$

e) $\frac{20m}{4m}$ **f)** $\frac{18xy}{3y}$ **g)** $\frac{10a^2}{5a}$ **h)** $\frac{24pq}{8pq}$

F Problems ●●●

6. Write an expression for each perimeter, then simplify it.

a) A triangle with sides $4x$, $5x$ and $6x$

b) A rectangle with length $8a$ and width $5a$

c) A rectangle with length $4x + 3$ and width $3x$

★ Extension

7. A rectangle is $2x$ long and $5x$ wide. Write a simplified expression for its area.

8. Simplify $5x + 3y - 2x + 4y - x - 5y$.

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